

Statement of Work

High-Performance Liquid Chromatograph-Mass Spectrometer (HPLC-MS)

6 January 2026

I. Background:

A new High-Performance Liquid Chromatograph-Mass Spectrometer (HPLC-MS) is needed. The current HPLC-MS (Agilent 1260 Infinity) is over 12 years old, the mass spectrum analyzer is inoperable, and the computer operating this instrument uses outdated software (i.e. Windows 7). HPLC-MS is an analytical chemistry technique used to study a wide range of organic chemicals present in water, including chemical warfare agents (e.g. VX nerve agent), pesticides, and many other toxic chemicals.

II. Minimum Technical Requirements:

The following specification lists the minimum (threshold) requirements for the HDR M700:

1. The HPLC-MS must be a component system.
 - a. The system Components must be from the same manufacturer.
 - b. All components must be serviced by the OEM manufacturer.
 - c. The components of this system must communicate using a controller area network (CAN).
2. The HPLC-MS must have an operating pressure of at least 800bar.
3. Mass Spectrum Analyzer must have two modes of operation Electro-spray ionization (ESI) and Atmospheric-pressure chemical ionization (APCI) with the following options:
 - a. Atmospheric-pressure chemical ionization (APCI) heats the buffer liquid to a temperature of at least 750°C aerosolized in nitrogen gas.
 - b. The mass to charge range (m/z) must be at least 2 to 3000 for ESI mode of operation.
 - c. The sensitivity must be less than 70 Femto-grams of material Reserpine using a single ion mass spectrometry (SIM) at 609.3 AMU with positive mode ESI.
 - d. Must have an autotune feature.
 - e. Must operate on 110v.
4. This mass spectrum analyzer must have a fixed nebulizer.
5. The system must have a Quaternary pump and a degasser.
 - a. This pump must have use buffers with an pH ranging from 1.0 to 12.5.
 - b. The flow rate of this pump must range from 0.001ml to 5ml.
 - c. The maximum operating pressure must be up to 800 bars (11601psi).
6. It must also include the brand name of better Mass Hunter Data Analysis Software.

7. The Control and Analysis software must be upgradable as software updates are released.
8. The HPLC-MS must have the following detector type:
 - a. include an Ultraviolet-Visible (UV-VIS) and a Diode Array Detector with a minimum of 60-millimeter path length.
 - b. Signal to noise ratio less than 0.6 μ AU/cm
 - c. Must and be able to maintain a flat baseline.
 - d. Have a minimum 1024-element photodiode internal detector.
 - e. Be able to measure a minimum of 8 signals simultaneously.
9. Must use deuterium lamp source for Ultra-Violet light detection
10. Must have an Autosampler with the following specifications:
 - a. cooled and humidity controlled to minimize sample decomposition
 - b. Operating temperature ranges of 4°C to 40°C.
 - c. An injection time of less than 10s.
 - d. Must have a multi-wash capability.
11. Heated and cooled column compartment that will hold at least 4 columns, and heat exchangers for solvent preheating and cooling.
 - a. The column compartment has two independent temperature zones that can cool 10°C less than ambient temperature to 85°C.
 - b. Maintain a stable temperature to 0.1°C.
12. Must have a column compartment with two independent temperature zones that can cool to a temperature of 10°C and heat to a temperature of 85°C
13. The column must maintain a table temperature of 0.1°C.
14. The system must be upgradable to supercritical fluids (SFC) of Food Grade carbon dioxide (CO₂).
15. Nitrogen Generator operating at either 110Volts or 208 Volts for Mass Spectrum Analyzer and to degas solvents.
16. This instrument must be purchased directly from the manufacturer.
17. Installation must be included, and installation must be performed by the OEM manufacturer.
18. Training for 3 people for 3 days to be included
19. Delivery: 6 months from receipt of order.

III. Qualifying Country Products:

All products proposed/delivered to satisfy these requirements shall be either domestic or qualifying country end products in accordance with DFARS 252.225-7001. Shall include all shipping fees, import or duty taxes and comply with Buy American and Balance of Payments Program.

IV. Delivery and Training:

- A. Delivery is required by the earliest possible delivery date, not to exceed 9 months after receipt of Award.
- B. Training must be performed by the OEM manufacturer for 3 people over 3 days.
- C. Delivery Address:

POC: Dr. Daniel Felker
Air Force Institute of Technology/ENP
2950 Hobson Way
WPAFB, OH 45433

- D. DoDAAC – F4F5AL

V. Period of Performance:

- 1. The vendor shall deliver and complete all requirements of this Statement of Work no later than 9 months after contract award.
- 2. Contractor shall be responsible for transportation of all new products, loading, unloading, storage and all necessary movement from factory to final onsite set up.
- 3. The contractor Project Manager shall be responsible for on-site coordination of all deliveries of their product, to include access to and operation of any equipment needed for unloading and movement of materials or products necessary to complete all requirements of this Statement of Work.
- 4. Free from defects in material and workmanship, under normal and proper use in accordance with equipment user manual, for the period of time equal to two years after receipt.

V. Delivery Procedures:

All vehicles larger than a large pick-up truck is required to be inspected by the Wright- Patterson Air Force Base Commercial Vehicle Delivery Gate (CVDG) prior to entering the installation.

1.0 Vehicles to be inspected include, but are not limited to, the following:

1.1 Step van/panel truck

1.2 Tractor/trailer, box, and flatbed containing cargo

1.3 Tanker trucks

1.4 Box Trucks

1.5 Tour buses

- 1.6 Garbage/recycled waste trucks
- 1.7 Concrete trucks/mixers, dump trucks
- 1.8 Cranes, recreational vehicles, petroleum tanker

2. This inspection will be conducted at Gate 16A located off State Route 444. The following are exceptions to vehicles utilizing the CVDG:

2.1 If the vehicle has the product inside (concrete and asphalt trucks) and timely delivery is necessary due to product deterioration it does not need to enter the CVDG. To bypass the CVDG, the contractor shall submit a list containing drivers' names, social security numbers and the state in which the driver's license is held for those drivers who will be entering the base. This shall be accomplished 24 hours prior to requested entry time. If entry is requested on Monday, this list must be submitted by Friday at 1630 hours. All lists shall be submitted to the 88th ABW/CE Directorate contract inspector. The only gates that may be used under this exemption shall be 15A, 26A, 38A, and gate 1B. If the driver's name is not on the list, he/she will not be allowed access to the installation through these gates and the base will not assume liability for denied access.

2.2 If a delivery vehicle must exit, and then re-enter the base to complete its route, the vehicle shall be resealed upon exiting the base. After initially passing through the commercial vehicle delivery gate, trucks shall be resealed at Gates 15A, 38A and 22B. The resealing of the trucks will allow them to continue to any other area of the installation (Areas A, B, or Kittyhawk) without reprocessing through the CVDG. To receive resealing assistance, the drivers shall physically stop at one of the three authorized gates and request the installation entry controller to reseal their truck and provide the next location of their delivery. The controller will reseal the truck and give the delivery driver a pre-clearance form. The driver shall present the pre-clearance form to the entry controller at the next point of installation entry. This reentry can be at any base gate.

2.3 Wright Patterson Integrated Defense Plan, paragraph 3.10.12.9.2. All irregular deliveries will be sent to the CVDG for inspection. All visitors, which include contractors and commercial vendors without proper background checks and contractor passes, will not be allowed on WPAFB unless escorted by the receiving agency. This will take place after the individual(s) are issued a pre-clearance form at the CVDG. After successful completion of the vehicle inspection, the receiving agency will escort the individual to and from their destination on WPAFB. An inspector at the CVDG will contact the receiving agency, informing them to send an escort to the CVDG. Escorts will remain with the vehicle at all times while it is on WPAFB.

2.4 Vehicles may be subject to an inspection at any of installation entry control points during a directed random antiterrorism measure (RAM). Any commercial

vehicle, regardless of size, can be directed to the CVDG at the discretion of the installation entry controller.

VI. Payment:

100% payment upon completion of all requirements specified in this Statement of Work. Payment shall be made on a net 30 basis. Invoices shall be submitted and paid through Wide Area Workflow. (<https://wawf.eb.mil>)